Vasicek vs CR+

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1 Introduction

The Vasicek model and CreditRisk+ (CR+) are two of the most widely used frameworks for modelling credit portfolio risk. Although their mathematical formulations differ substantially, both models are designed to describe the same economic phenomenon: the impact of systematic credit risk on a large portfolio of obligors. In both frameworks, defaults become correlated because borrowers are exposed to a common risk driver representing the state of the economy. In the asymptotic limit of a large homogeneous portfolio, idiosyncratic risk is diversified away and portfolio losses are entirely determined by this common factor. Consequently, both models can be viewed as asymptotic single-factor credit portfolio models. In the large portfolio limit, portfolio losses are driven primarily by a single systematic risk factor. The principal difference between the models lies in how the systematic risk factor is represented:

- In the Vasicek model, systematic risk is described by a latent normally distributed factor Z and an asset correlation parameter ρ.
- In CreditRisk+, systematic risk is represented by a stochastic default intensity Λ, typically assumed to follow a gamma distribution.

Despite these differing representations, the models can be calibrated so that their loss distributions become nearly identical in the large portfolio limit. This can be achieved either by matching the variance of the common risk driver or by matching a specific quantile of the loss distribution, such as the 99% Value-at-Risk. Both approaches are discussed below.

2 Systematic Risk in Vasicek and CreditRisk+

Both models generate default dependence through a portfolio-wide systematic state variable. The specific mathematical constructions differ, but the economic interpretation is the same: favourable systematic states produce few defaults, while adverse states produce large clusters of defaults.

2.1 Vasicek Model

In the Vasicek model, the credit quality of borrower i is represented by the latent variable

$$X_i = \sqrt{\rho}\, Z + \sqrt{1-\rho}\, \varepsilon_i,$$

where

$$Z \sim N(0,1), \qquad \varepsilon_i \sim N(0,1),$$

and the idiosyncratic variables ε_i are mutually independent and independent of Z. A default occurs when

$$X_i < \Phi^{-1}(p),$$

where p denotes the unconditional probability of default. Conditioning on a realization $Z = z$ gives

$$P(X_i < \Phi^{-1}(p) \mid Z = z) = P\Big(\sqrt{\rho}\, z + \sqrt{1-\rho}\, \varepsilon_i < \Phi^{-1}(p)\Big).$$

Rearranging yields

$$P\left(\varepsilon_i < \frac{\Phi^{-1}(p) - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right).$$

Since ε_i is standard normally distributed,

$$p(z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right).$$

This conditional probability of default is the key quantity governing portfolio losses in the asymptotic limit. A large positive value of z corresponds to a strong economy and a low conditional default rate, whereas a large negative value of z produces elevated default probabilities and clustered losses.

2.2 CreditRisk+

CreditRisk+ introduces systematic dependence through a stochastic default intensity. Let N denote the number of defaults in a portfolio during a given time horizon. Conditional on a realization of the intensity factor $\Lambda = \lambda$, defaults are assumed to occur according to a Poisson distribution,

$$N \mid \Lambda = \lambda \sim \text{Poisson}(\lambda).$$

It follows that

$$E[N \mid \Lambda = \lambda] = \lambda, \qquad \text{Var}(N \mid \Lambda = \lambda) = \lambda.$$

Systematic risk is introduced by allowing the default intensity itself to vary across economic states. In the standard CreditRisk+ formulation,

$$\Lambda \sim \text{Gamma}(\alpha, \beta),$$

where α denotes the shape parameter and β the scale parameter. The moments of the gamma distribution are

$$E[\Lambda] = \alpha\beta,$$

and

$$\text{Var}(\Lambda) = \alpha\beta^2.$$

The mean determines the unconditional expected number of defaults, while the variance determines the degree of systematic uncertainty around that expectation. A useful measure of the variability of the common risk factor is the squared coefficient of variation,

$$CV^2 = \frac{\text{Var}(\Lambda)}{E[\Lambda]^2}.$$

Substituting the moments of the gamma distribution gives

$$CV^2 = \frac{\alpha\beta^2}{(\alpha\beta)^2},$$

which simplifies to

$$CV^2 = \frac{1}{\alpha}.$$

Hence, the shape parameter completely determines the relative variability of the common intensity factor. A low value of α corresponds to a highly volatile credit environment, whereas a large value of α produces a more stable environment with weaker systematic fluctuations. The economic interpretation is analogous to the Vasicek model. Different realizations of Λ correspond to different systematic states of the economy: a low value of Λ implies a benign credit environment with relatively few defaults, while a high value of Λ corresponds to stressed conditions with elevated default rates throughout the portfolio.

2.3 Common Interpretation

The two models can therefore be viewed as alternative representations of the same underlying mechanism. In the Vasicek model, the common state variable is the Gaussian factor Z, which determines the conditional probability of default

$$p(Z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, Z}{\sqrt{1-\rho}}\right).$$

In CreditRisk+, the common state variable is the gamma-distributed intensity

$$\Lambda,$$

which directly determines the conditional default frequency. In the large portfolio limit, portfolio losses are fully determined by the realization of these common factors. The asymptotic loss rate may therefore be written schematically as

$$L_\infty = p(Z)$$

in the Vasicek model and

$$L_\infty \propto \Lambda$$

in CreditRisk+. To illustrate the similarity, consider a portfolio of 10 000 identical loans with an unconditional probability of default of 1%. The expected number of defaults is therefore 100 per year. In CreditRisk+, different realizations of the common intensity correspond to different economic states:

- A favourable state with a low realization of Λ, producing substantially fewer than 100 defaults.
- An average state with $\Lambda \approx E[\Lambda]$, producing approximately 100 defaults.
- An adverse state with a high realization of Λ, producing a large cluster of defaults.

Exactly the same interpretation can be made in the Vasicek model. Once the systematic factor takes a realized value $Z = z$, all borrowers share the same conditional probability of default

$$p(z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right).$$

The realization of z therefore determines the overall state of the credit portfolio:

- A favourable state corresponds to a high value of z, for which $p(z)$ becomes substantially lower than 1% and only a small number of loans default.
- An average state corresponds approximately to $z \approx 0$, for which the default rate is close to its unconditional level and the number of defaults is around 100.
- An adverse state corresponds to a low value of z, for which $p(z)$ increases significantly and a large cluster of defaults occurs.

In both models, defaults are conditionally independent once the common risk factor has been fixed. In CreditRisk+, this factor is the stochastic intensity Λ, while in the Vasicek model it is the Gaussian factor Z. Default clustering therefore does not arise from direct interactions between obligors, but from their shared exposure to the same systematic state of the economy. From this perspective, the principal difference between the models is not the economic mechanism but the mathematical representation of the systematic risk driver. Vasicek models the state of the economy through a normally distributed latent factor, whereas CreditRisk+ models it through a gamma-distributed default intensity. Both approaches generate a single-factor loss distribution whose shape is determined by the variability of the common systematic risk factor.

3 Approximate Mapping between Vasicek Asset Correlation and CreditRisk+ Gamma Volatility

Although the Vasicek model and CreditRisk+ are based on different mathematical frameworks, both separate credit risk into an idiosyncratic component and a systematic component driven by a common economic factor. In the Vasicek model, this decomposition is explicit in the latent asset variable. In CreditRisk+, it appears through the decomposition of default-count variance into ordinary Poisson variation and variation in a stochastic default intensity. This common structure makes it possible to compare the two models by comparing the relative importance of systematic and idiosyncratic risk. Doing so leads to a useful approximation linking the Vasicek asset correlation ρ to the CreditRisk+ gamma volatility.

3.1 Systematic and idiosyncratic risk in the Vasicek model

In the Vasicek model, the latent variable of obligor i is

$$X_i = \sqrt{\rho}\, Z + \sqrt{1-\rho}\, \varepsilon_i,$$

where both Z and ε_i are standard normally distributed and mutually independent. Since

$$\mathrm{Var}(X_i) = \rho + (1-\rho) = 1,$$

the total variance can be decomposed into

$$\underbrace{\rho}_{\text{systematic risk}} + \underbrace{(1-\rho)}_{\text{idiosyncratic risk}}.$$

The parameter ρ therefore represents the fraction of total asset variance attributable to the common systematic factor, while $(1-\rho)$ represents the fraction attributable to borrower-specific effects. A natural measure of the relative importance of systematic risk is therefore

$$\frac{\text{systematic risk}}{\text{idiosyncratic risk}} = \frac{\rho}{1-\rho}.$$

When ρ is small, defaults are driven primarily by borrower-specific events and portfolio losses exhibit little dependence on the economic environment. As ρ increases, the common factor becomes more important and defaults become increasingly clustered during adverse economic conditions. In the asymptotic portfolio limit, idiosyncratic fluctuations are diversified away and portfolio losses become entirely driven by the realization of the common factor Z. The ratio

$$\frac{\rho}{1-\rho}$$

can therefore be viewed as a natural measure of the strength of systematic credit risk in the Vasicek framework.

3.2 Systematic and idiosyncratic risk in CreditRisk+

In CreditRisk+, the number of defaults is assumed to be conditionally Poisson distributed,

$$N \mid \Lambda \sim \text{Poisson}(\Lambda),$$

where the default intensity Λ follows a gamma distribution. Applying the law of total variance gives

$$\text{Var}(N) = E[\text{Var}(N \mid \Lambda)] + \text{Var}(E[N \mid \Lambda]).$$

Since

$$E[N \mid \Lambda] = \Lambda, \qquad \text{Var}(N \mid \Lambda) = \Lambda,$$

it follows that

$$\text{Var}(N) = E[\Lambda] + \text{Var}(\Lambda).$$

This decomposition can be interpreted as

$$\text{Var}(N) = \underbrace{E[\Lambda]}_{\text{idiosyncratic risk}} + \underbrace{\text{Var}(\Lambda)}_{\text{systematic risk}}.$$

The first term represents ordinary Poisson variation and corresponds to borrower-specific randomness. The second term captures fluctuations in the common default intensity and therefore represents systematic risk. By analogy with the Vasicek model, a natural measure of the relative importance of systematic risk would therefore be

$$\frac{\text{systematic risk}}{\text{idiosyncratic risk}} = \frac{\text{Var}(\Lambda)}{E[\Lambda]}.$$

However, this quantity depends on the scale of the portfolio through $E[\Lambda]$. Doubling the portfolio size doubles the expected number of defaults and changes the ratio, even if the underlying systematic risk remains unchanged. A more useful measure is therefore obtained by normalizing the stochastic intensity by its mean. For a gamma-distributed intensity,

$$CV^2 = \frac{\text{Var}(\Lambda)}{E[\Lambda]^2},$$

where CV denotes the coefficient of variation. The quantity CV^2 is dimensionless and independent of portfolio size. It measures the variability of the common credit environment relative to its average level. A larger value of CV^2 implies larger fluctuations in the common default intensity and consequently stronger dependence between obligors. In this sense, CV^2 plays a role analogous to

$$\frac{\rho}{1-\rho},$$

which measures the relative importance of the common systematic factor in the Vasicek model.

3.3 Approximate parameter mapping

Both models can therefore be interpreted as decomposing credit risk into an idiosyncratic component and a systematic component. In the Vasicek model, the relative importance of these components is measured by

$$\frac{\rho}{1-\rho},$$

whereas in CreditRisk+ the variability of the common risk factor is measured by

$$CV^2.$$

Identifying these measures of systematic risk gives the approximation

$$CV^2 \approx \frac{\rho}{1-\rho}.$$

Solving for ρ yields

$$\rho \approx \frac{CV^2}{1+CV^2}.$$

This relationship provides a practical heuristic mapping between the CreditRisk+ gamma volatility and the Vasicek asset correlation. While not exact, it captures the idea that both parameters quantify the strength of systematic credit risk relative to idiosyncratic risk.

4 Calibrations

In this section, we consider two common calibration approaches linking the Vasicek model and CreditRisk+. The first approach matches the variance of the common risk driver, while the second matches a tail quantile such as the 99% Value-at-Risk.

4.1 Calibration via gamma variance and asset correlation

Assume that the asset correlation in the Vasicek model is $\rho = 10\%$. At a 99% confidence level, the outcome corresponds to a strongly negative realization of the systematic factor Z. The conditional PD then becomes significantly higher than 1%, which leads to a large cluster of defaults. By exploiting the relationship derived earlier, we translate this into CreditRisk+ using

$$CV^2 \approx \frac{\rho}{1-\rho}.$$

In the numerical example,

$$\rho = 0.10.$$

Inserted into the expression above yields

$$CV^2 \approx \frac{0.10}{1-0.10} = \frac{0.10}{0.90} \approx 0.111.$$

By definition, the coefficient of variation satisfies

$$CV = \frac{\sqrt{\mathrm{Var}(\Lambda)}}{E[\Lambda]}.$$

Thus,

$$CV = \sqrt{CV^2} = \sqrt{0.111} \approx 0.333.$$

Economic interpretation This implies that the standard deviation of the common default intensity Λ is approximately 33% of its mean. At the 99% quantile, a high value of Λ is realized, leading to essentially the same number of defaults as in the Vasicek model. Numerical comparisons typically show that the 99% VaR differs only marginally between the two models.

4.2 Calibration of ρ to a Given VaR Level

In practical applications, it may be desirable not only to match the variance of the common risk driver but also to match a specific quantile of the loss distribution. This is particularly relevant when CreditRisk+ is used as the primary model and one wishes to determine the corresponding asset correlation in the Vasicek framework. Assume that CreditRisk+ produces a 99% loss level (VaR) of

$$L_{0.99}^{CR+} = \ell_{99}.$$

We seek the value of ρ such that the Vasicek model produces the same portfolio loss at the 99% confidence level:

$$L_{0.99}^{Vasicek}(\rho) = \ell_{99}.$$

Portfolio loss in the Vasicek model For a homogeneous portfolio with identical exposures and constant loss given default (LGD), the asymptotic portfolio loss conditional on the systematic factor realization $Z = z$ is

$$L(z) = \text{LGD} \cdot p(z),$$

where

$$p(z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right)$$

is the conditional probability of default. To simplify the notation, we set

$$\text{LGD} = 1$$

in the following. The derivation is unchanged for any other constant LGD, since the loss distribution is then obtained by a simple rescaling. Under this normalization,

$$L(z) = p(z).$$

The 99% loss level Since portfolio losses are a decreasing function of the systematic factor Z, the 99% loss quantile corresponds to the 1% lower-tail realization of Z,

$$z_{0.01} = \Phi^{-1}(0.01).$$

The Vasicek 99%-VaR is therefore

$$L_{0.99}^{Vasicek} = p(z_{0.01}),$$

which yields

$$L_{0.99}^{Vasicek} = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, \Phi^{-1}(0.01)}{\sqrt{1-\rho}}\right).$$

Using the symmetry property

$$\Phi^{-1}(0.01) = -\Phi^{-1}(0.99),$$

we obtain the familiar ASRF representation

$$L_{0.99}^{Vasicek} = \Phi\left(\frac{\Phi^{-1}(p) + \sqrt{\rho}\, \Phi^{-1}(0.99)}{\sqrt{1-\rho}}\right).$$

Solution for ρ To match the CreditRisk+ VaR, we solve

$$\Phi\left(\frac{\Phi^{-1}(p) + \sqrt{\rho}\,\Phi^{-1}(0.99)}{\sqrt{1-\rho}}\right) = \ell_{99}.$$

Applying the inverse standard normal distribution function to both sides gives

$$\frac{\Phi^{-1}(p) + \sqrt{\rho}\,\Phi^{-1}(0.99)}{\sqrt{1-\rho}} = \Phi^{-1}(\ell_{99}).$$

Rearranging,

$$\sqrt{\rho}\,\Phi^{-1}(0.99) = \Phi^{-1}(\ell_{99})\sqrt{1-\rho} - \Phi^{-1}(p).$$

This equation can be solved analytically or numerically for ρ. Since the Vasicek VaR is a monotonic increasing function of the asset correlation, the solution is unique. For a constant LGD different from unity, the calibration condition becomes

$$\mathrm{LGD} \cdot L_{0.99}^{Vasicek} = \ell_{99},$$

or equivalently

$$L_{0.99}^{Vasicek} = \frac{\ell_{99}}{\mathrm{LGD}}.$$

Thus, a constant LGD merely rescales the loss distribution and does not affect the underlying relationship between VaR and the asset correlation.

5 Conclusion

The Vasicek model and CreditRisk+ provide alternative representations of systematic credit risk in large portfolios. Although their mathematical formulations differ, both describe default clustering through exposure to a common economic risk factor. By comparing the relative importance of systematic and idiosyncratic risk in the two frameworks, an approximate mapping can be established between the Vasicek asset correlation and the CreditRisk+ gamma volatility,

$$\rho \approx \frac{CV^2}{1 + CV^2}.$$

This relationship offers a practical way to translate model parameters between the two approaches. With appropriate calibration, the models can produce very similar portfolio loss distributions in large and well-diversified portfolios, particularly in the tail region relevant for risk measurement and capital assessment.